Chlorophyll-a Gap-Filling Using U-Net

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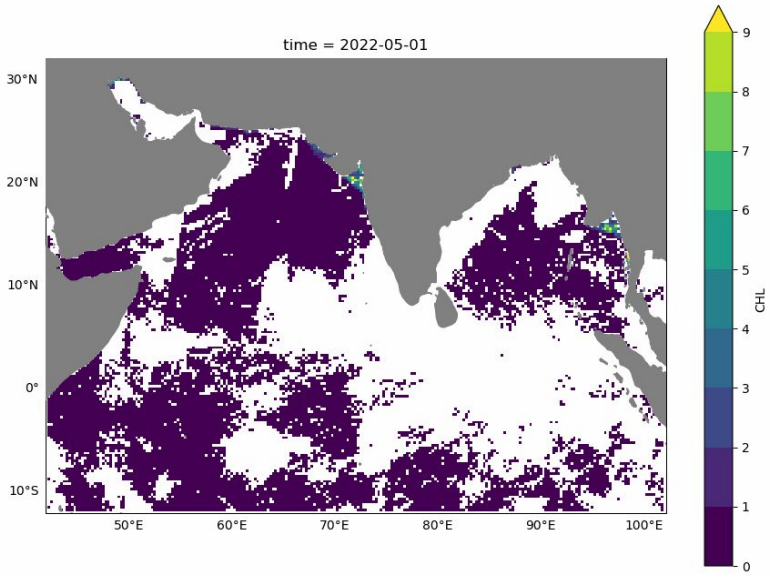
Problem: Missing CHL data

Satellite remote sensing: unable to penetrate through clouds

Traditional solution: interpolation & geospatial

Our solution: machine learning (neural networks)

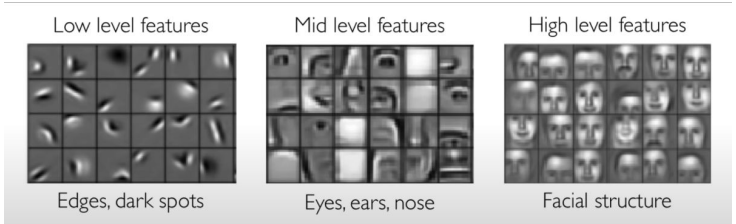
Goal: train a neural network with comparable performance, explore the benefit of incorporating environmental variables

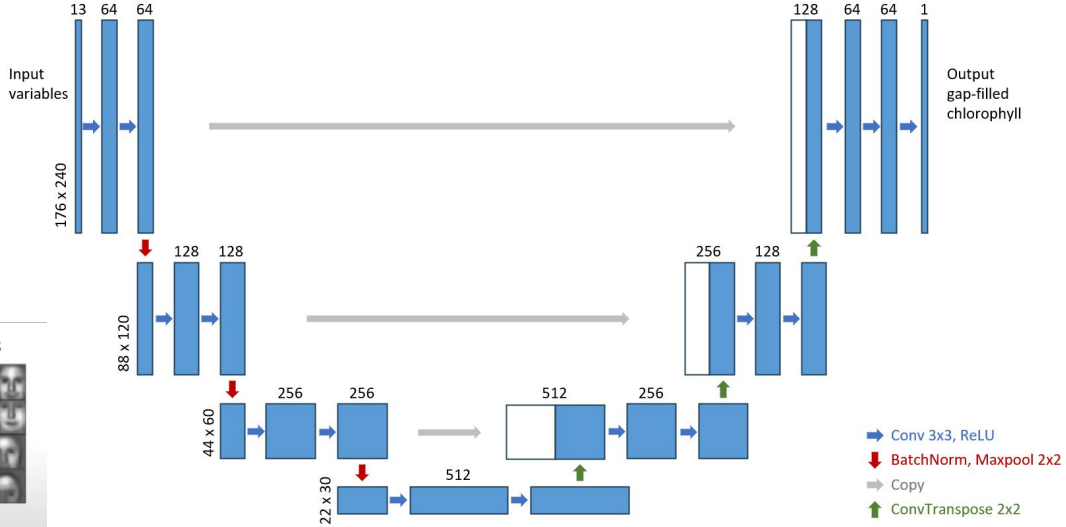


U-Net: A Convolutional Neural Network Architecture

Encoder: feature extraction with convolutional layers

Decoder: construct gap-filled image with extracted features





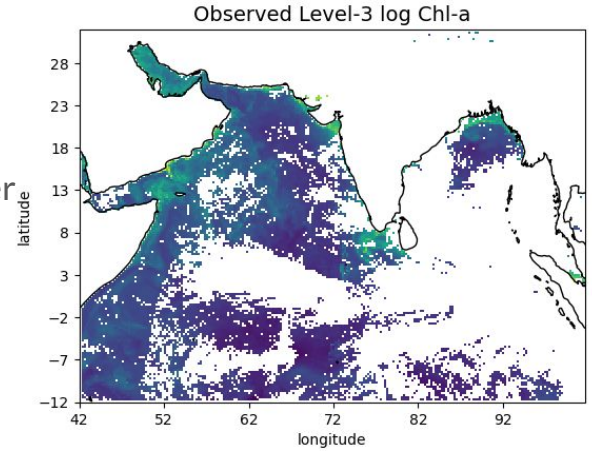
Variables & Data Processing

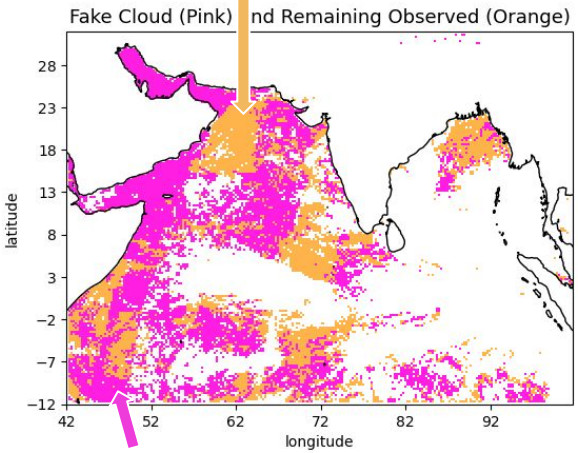
Data processing: mask observed chlorophyll with artificial clouds; log CHL; standardize; fill N/A with 0

Environmental variables:
SST, air temp, winds, time

CHL info: current, prior day, latter day masked CHL

Flags: binary flags for land, fake cloud, real cloud, observed CHL

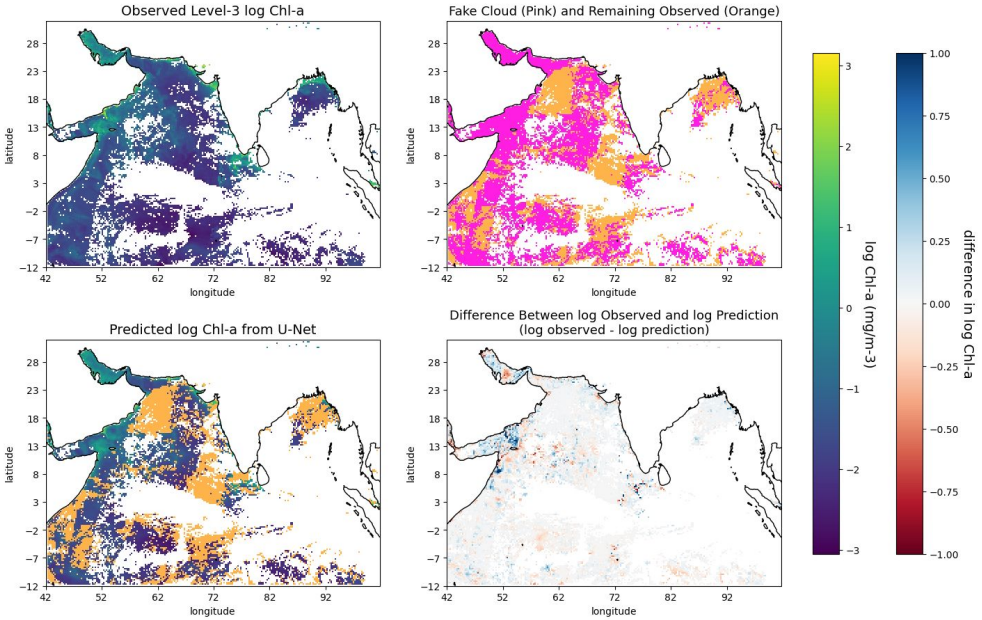




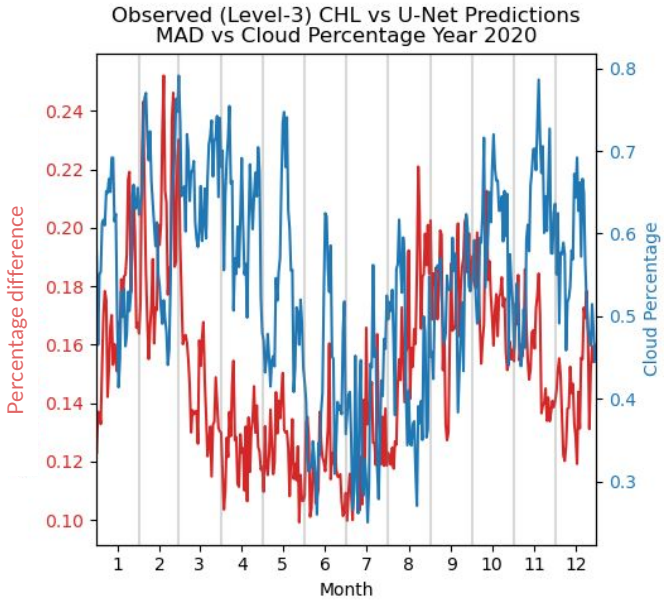
Remaining observed in which the predictions were based on

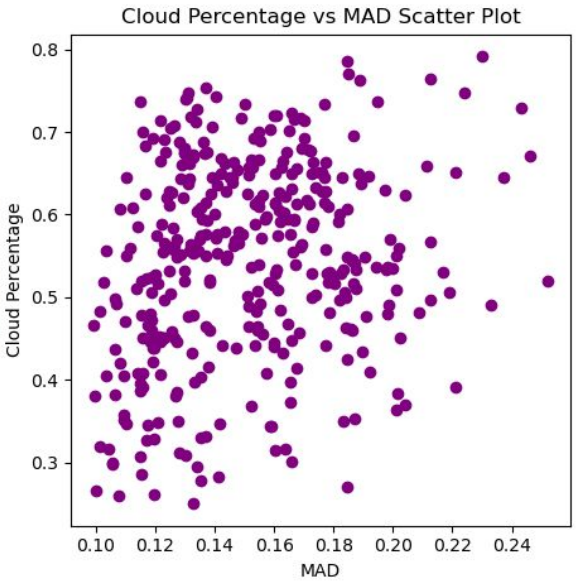
Evaluate prediction against the masked observed

Testing Against Observed - Visual Example

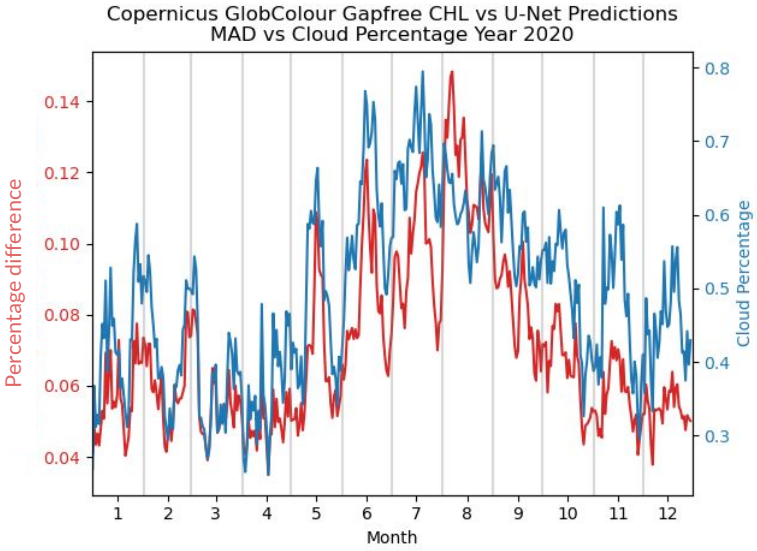


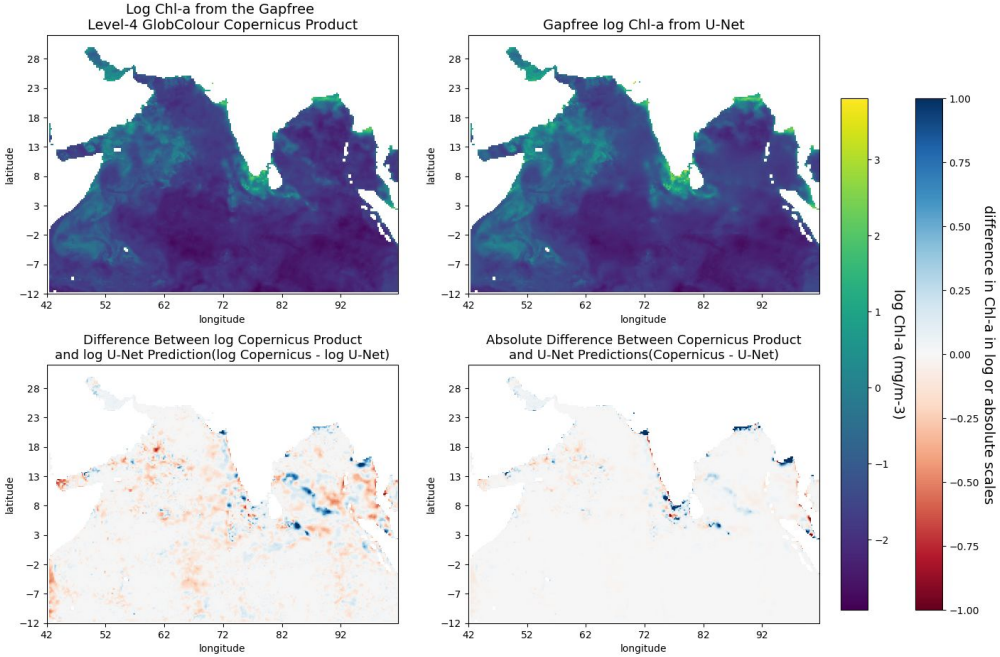
Testing Against Observed - Daily Mean Difference





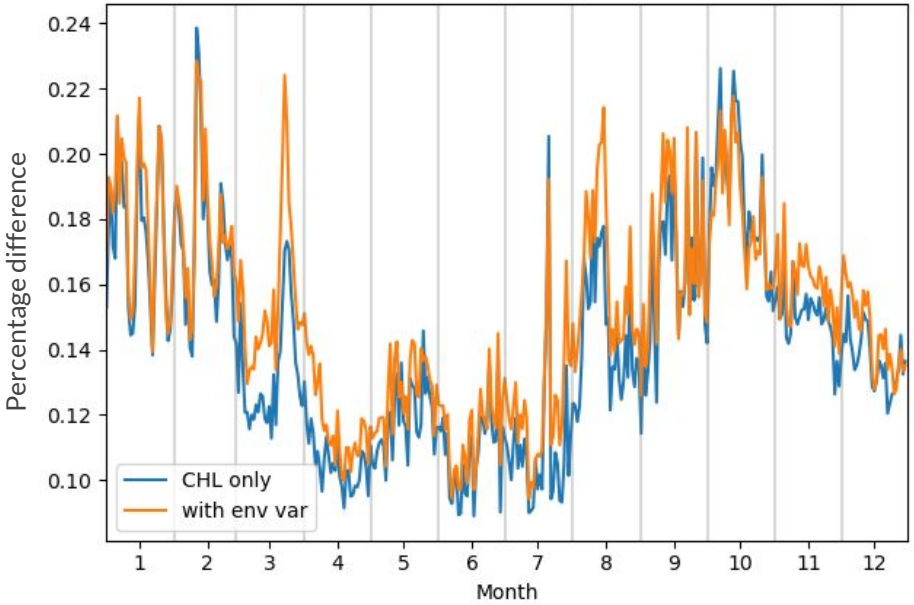
Compare Against Copernicus-GlobColour L4





Does Environmental Variables Help?

Sadly, not really :(

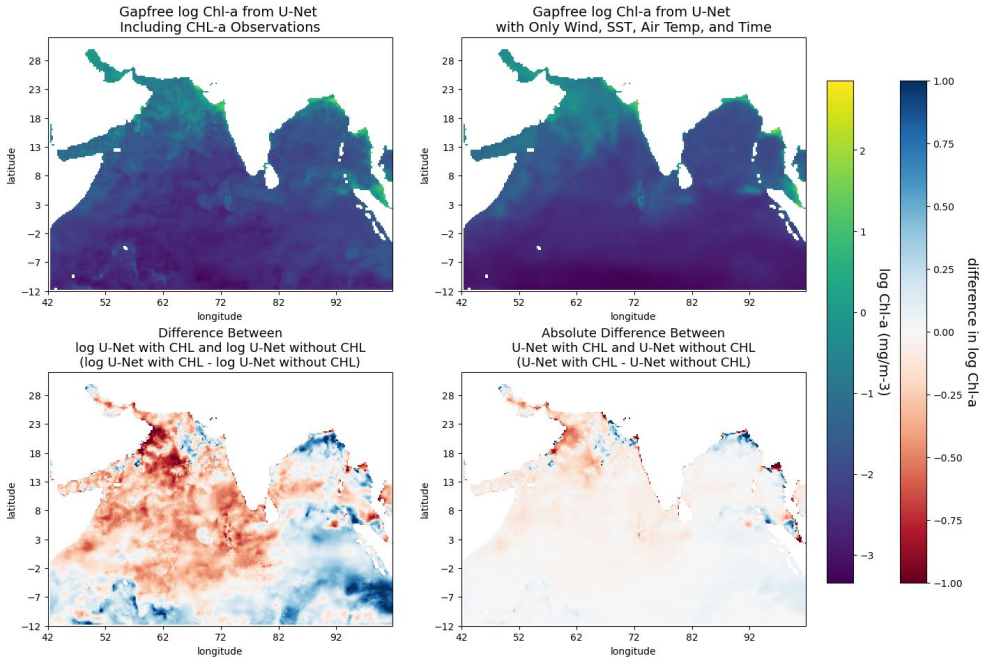


Are Environmental Variables Useful At All?

Yes!

Captures the general pattern

CAN reconstruct some details in certain regions



Main Takeaways

Initial model: work well, low mean absolute difference against the truth and similar to Level-4 product

Main difference with the Copernicus L-4 Gapfree product: lower coastline estimation

Environmental variables effectiveness: introduces noise with less missing chlorophyll; CAN capture chlorophyll patterns